



CJ 540

Sampling

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Population

All units of analysis (cases, individuals, groups, objects, events) that exist for a specific topic

Sample

A subset of the units of analysis in the population

Why do we sample?

Why not just take the population?



Why do we sample?

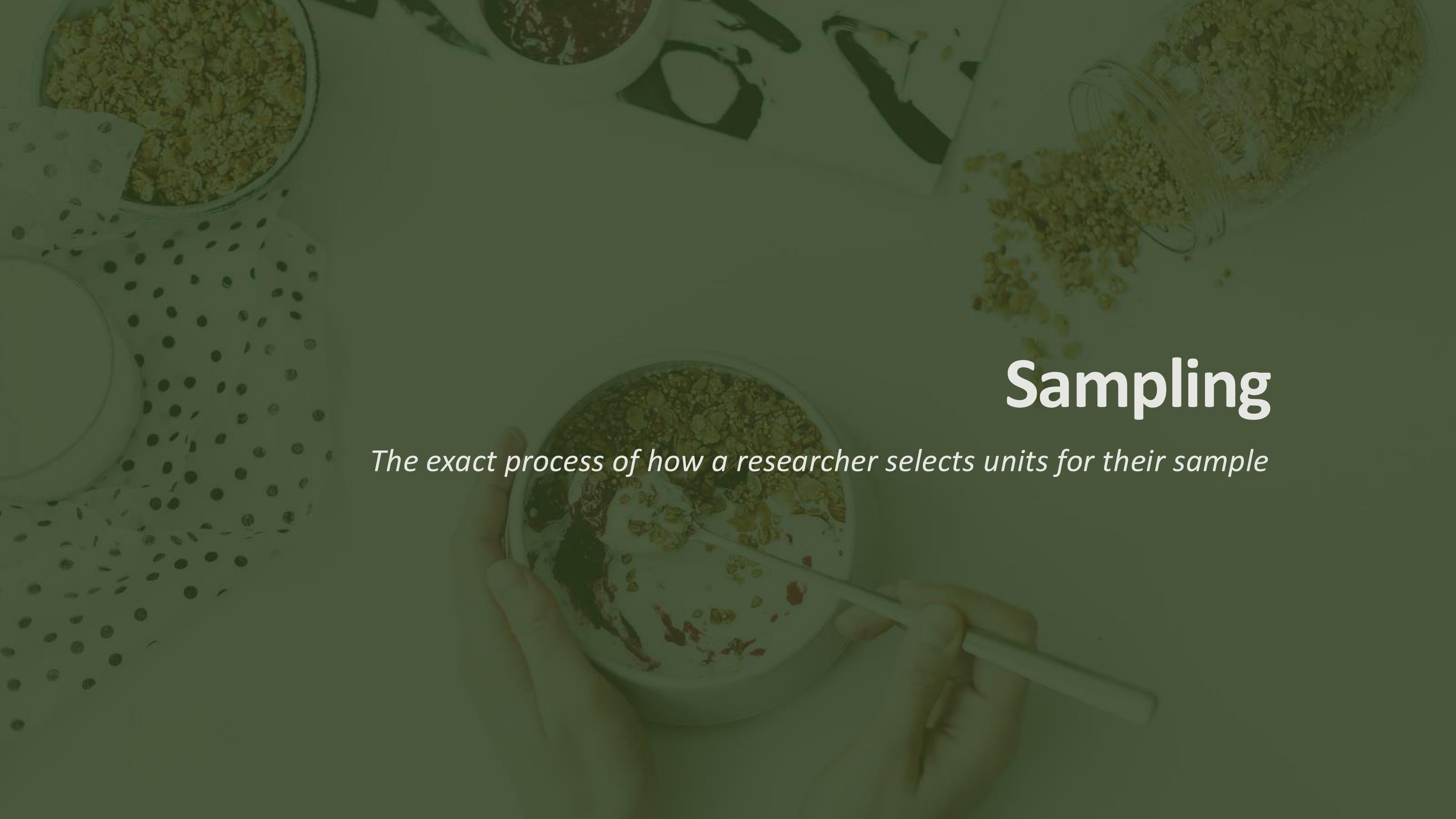
POPULATION REASONS

Time consuming to observe/collect data

Expensive

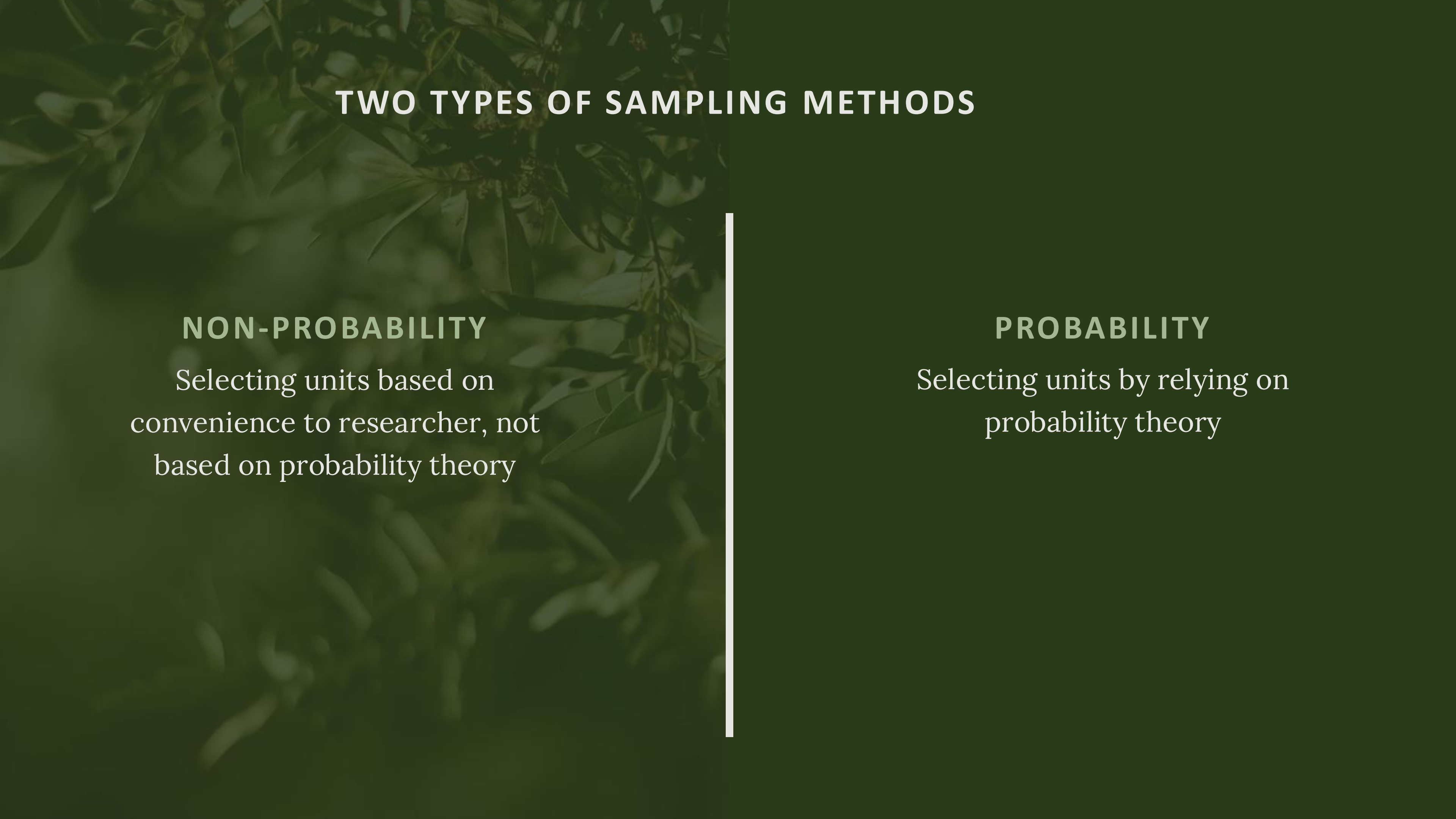
Population changes over time

Generalizability

A top-down view of a person's hands holding a white bowl of cereal. The person is using a white spoon to scoop up a portion of the cereal, which includes milk, granola, and some red fruit. The bowl is surrounded by various cereal-related items: a large bowl of granola in the top left, a small bowl of red fruit in the top center, a clear plastic container of granola in the top right, and a white bowl with a polka-dot pattern in the bottom left. The background is a light-colored surface.

Sampling

The exact process of how a researcher selects units for their sample



TWO TYPES OF SAMPLING METHODS

NON-PROBABILITY

Selecting units based on convenience to researcher, not based on probability theory

PROBABILITY

Selecting units by relying on probability theory



**Which method do we
use, and when?**

USES FOR EACH SAMPLING METHOD

NON-PROBABILITY

When researcher wants a non-representative sample (one that is not representative of the population)

Non-generalizable

Cannot provide exact estimates of a unit's likelihood of inclusion/exclusion

PROBABILITY

When researcher wants a sample that is representative of the population

Generalizable

Can provide exact estimates of a unit's likelihood of inclusion/exclusion

Generalizability

making claims about the population based on the sample

Representativeness

Sample has the same distribution of characteristics as the population from which it was selected.

A sample is representative of the population if all units within the population have an equal chance of being selected for the sample. Can make generalizable claims.

Bias

Sample bias/error is minimized if it looks like population.
Claims can be generalizable to population



POPULATION



SAMPLE



Non-Probability

Non-representative of
population



Probability

Representative of
population



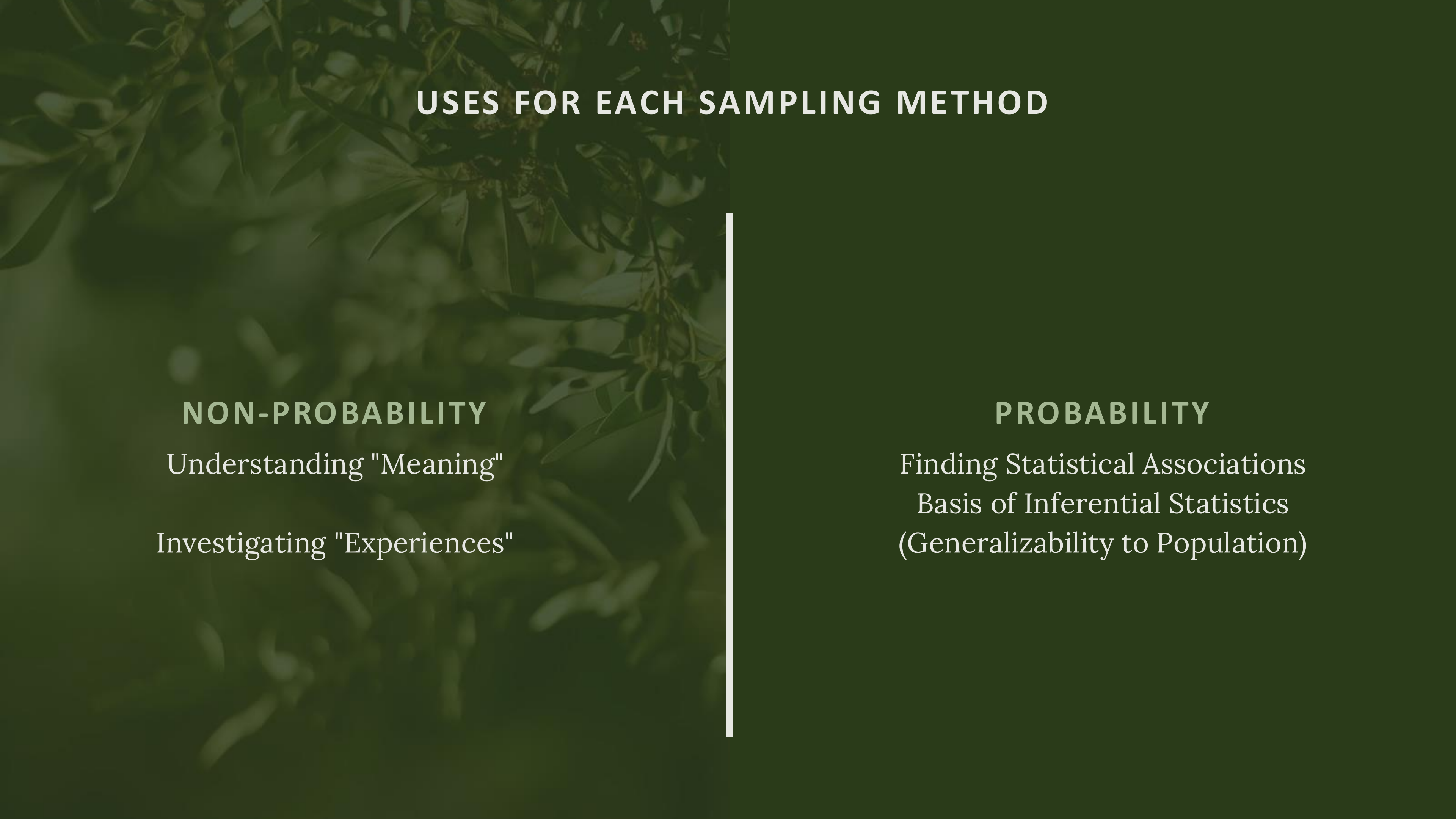
USES FOR EACH SAMPLING METHOD

NON-PROBABILITY

Qualitative Research

PROBABILITY

Quantitative Research



USES FOR EACH SAMPLING METHOD

NON-PROBABILITY

Understanding "Meaning"

Investigating "Experiences"

PROBABILITY

Finding Statistical Associations

Basis of Inferential Statistics
(Generalizability to Population)



Specific Sampling Methods by Type

NON-PROBABILITY

- Available subjects
- Snowball sampling
- Purposive sampling
- Quota sampling

PROBABILITY

- Simple Random Sampling
- Systematic Sampling
- Stratified Sampling
- ... etc

Non-Probability Sampling

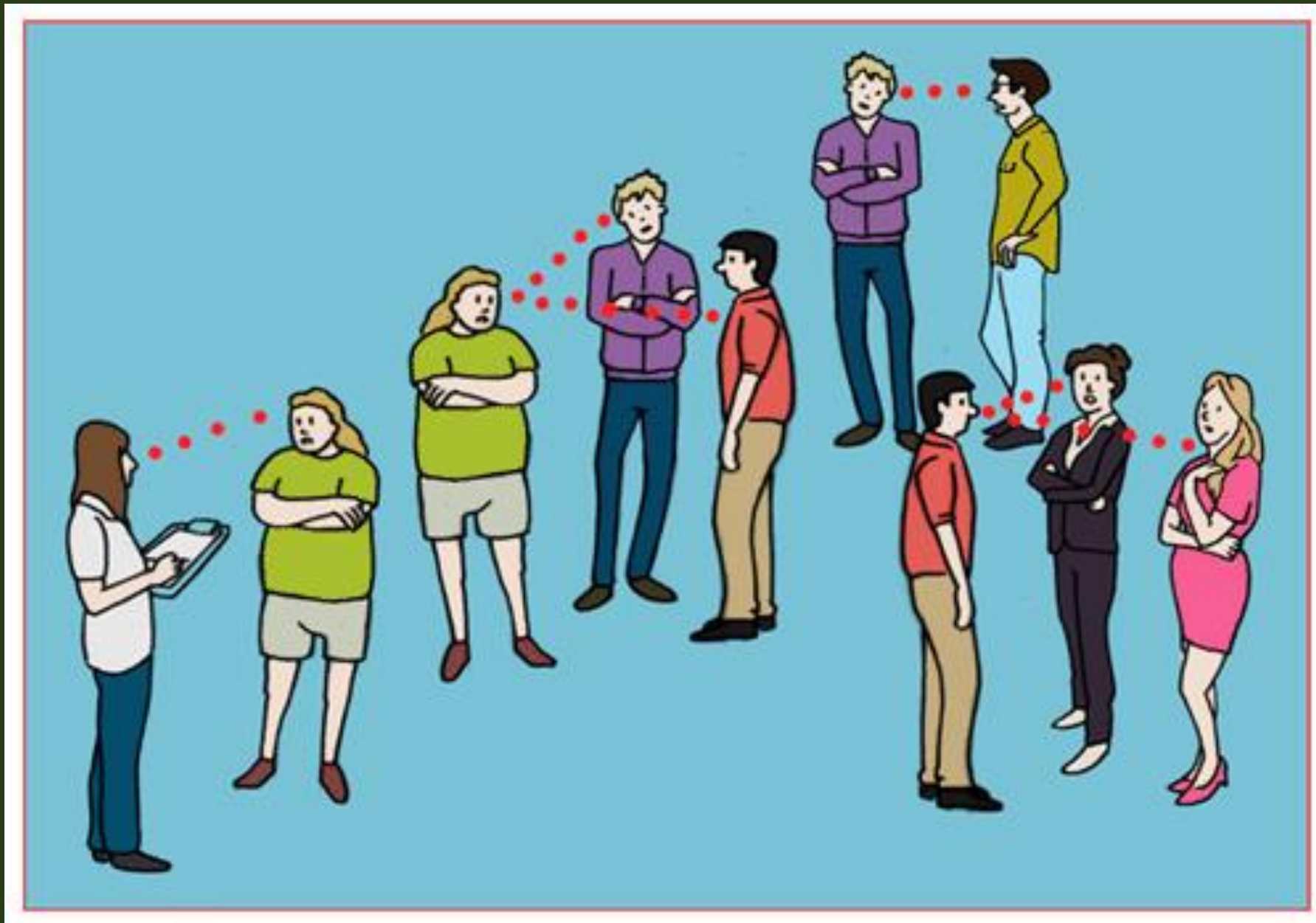
best for non-generalizable research

often used in qualitative research



AVAILABLE SUBJECTS

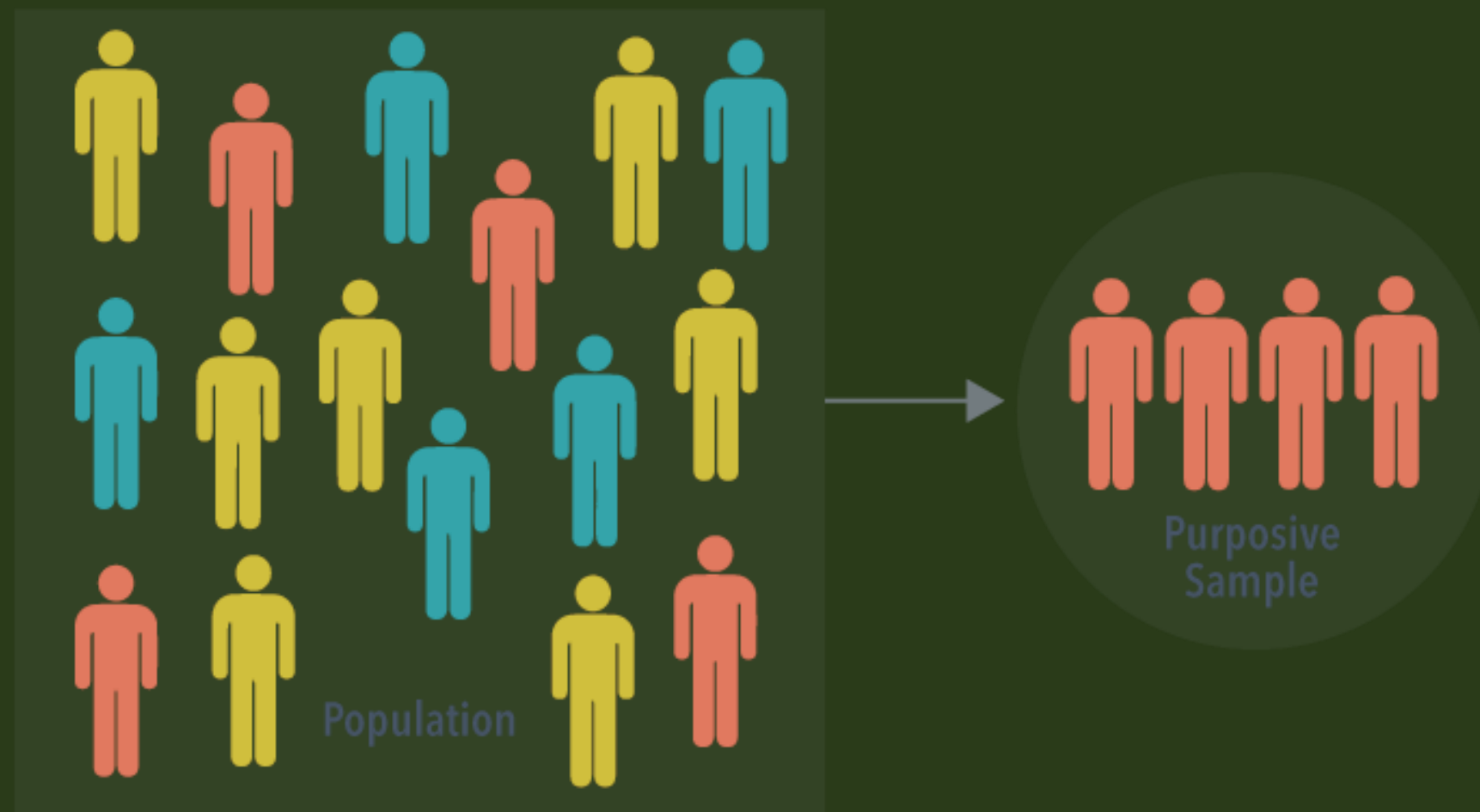
*units selected based on
convenience/proximity to researcher*



SNOWBALL SAMPLING

Each person interviewed is asked to suggest additional people for interviewing.

Purposive Sampling



PURPOSIVE SAMPLING

Units selected based on researcher's judgment about whether or not subjects are useful for findings/results.



QUOTA SAMPLING

Units selected based on pre-specified characteristics. Sample will have the same distribution of characteristics as the population

Things to Consider

Selecting Informants

Someone who is well versed in the topic (e.g. specific group) you wish to study, who is willing to tell you what they know, who can talk directly about the group/people and provide information, or who can introduce you to the group/people.

Gaining Entrée (Entry/Access to Group or Topic)

Meeting the gatekeepers (informants) of the group/topic and developing personal relationships with them, which can serve as a springboard for them trusting you as a researcher and (perhaps) make them more likely to participate or introduce you to other potential participants

Probability Sampling

best for generalizable, representative research (making statements about population)

necessary for quantitative research

can be used for qualitative research

How to we ensure representativeness and generalizability?

How to we abide by logic of probability theory?



Probability Sampling Requires

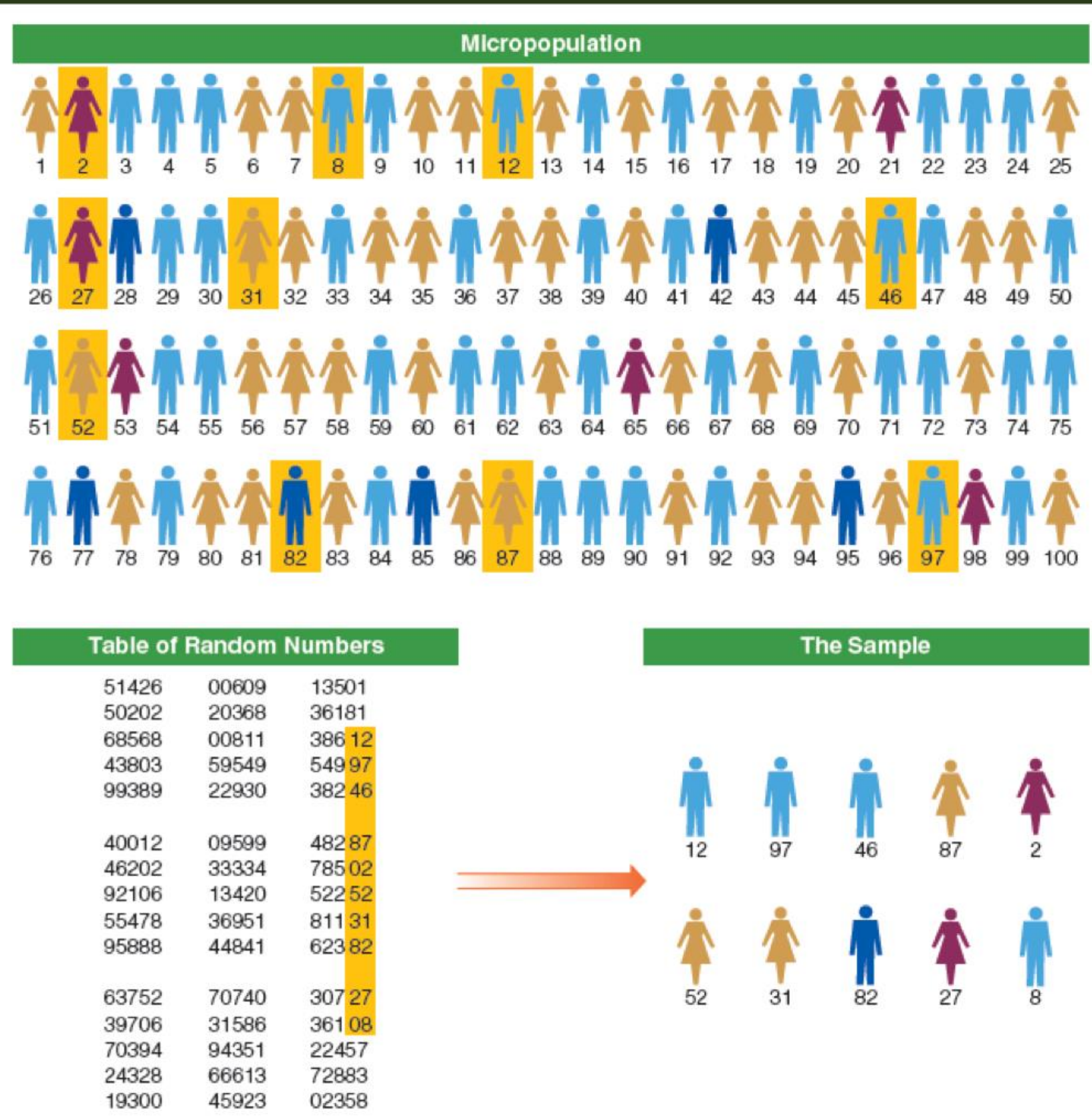
Random Selection

Each unit within the population has an equal chance of selection for the sample. They will be randomly selected from the population

Sampling Frame

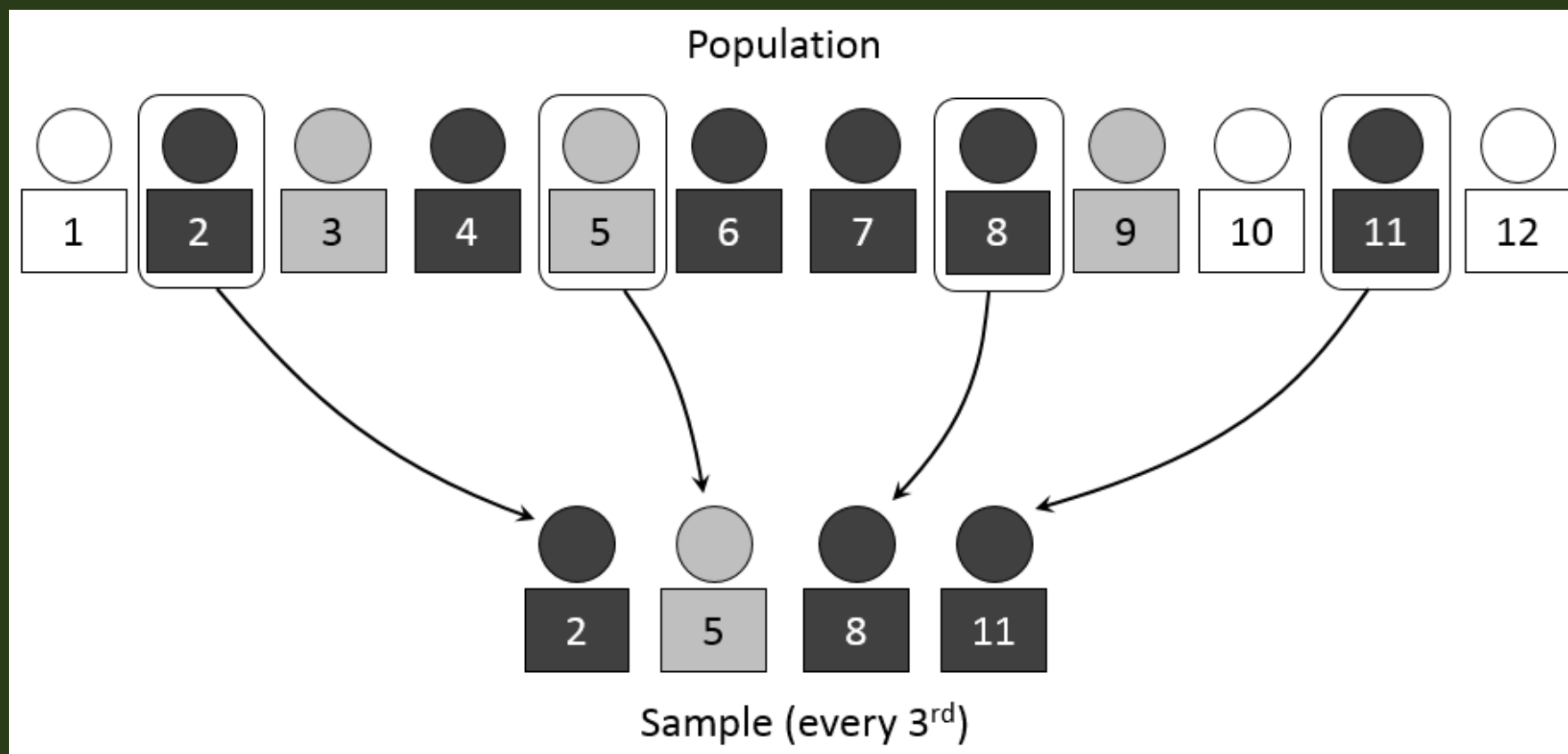
A list of all units in a population. Used to select which units will be in the sample. Note: Think about the structure of the sampling frame (randomize it).

All probability sampling needs a sampling frame



SIMPLE RANDOM SAMPLING

Units are assigned numbers. A set of random numbers is generated and the units having those numbers are included in the sample.



SYSTEMATIC SAMPLING

Every k th unit in a list is selected for inclusion in the sample.



STRATIFIED SAMPLING

Grouping units on sampling frame into homogenous groups (strata). Conduct systematic sampling on the entire list.